Oct 11. Separable simplicial algebras

Joint work with X. Huang, G. Sun and J. Zhang.

Introduction.

Let $\mathbb{K}$ be a commutative ring.

Δ be the category consisting of

objects = finite totally ordered sets

$[n] = \{0 < 1 < \cdots < n\}$.

morphism = order-preserving functions.

Def: A simplicial object X in a category $\mathcal{C}$ is a functor $X : \Delta^{op} \to \mathcal{C}$.

Equivalently. X consists of a collection of objects $\{X_n := X[n]\}_{n \geq 0}$ in $\mathcal{C}$. and a collection of morphisms in $\mathcal{C}$. for $0 \leq i \leq n$.

face maps $d_i = d_i^n = X_n \to X_{n-1}$

degeneracy maps: $s_i = s_i^n = X_n \to X_{n+1}$

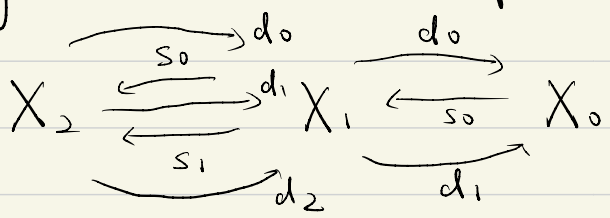
satisfying the simplicial identities

$$d_i d_j = d_{j-1} d_i \qquad i < j$$

$$s_i s_j = s_{j+1} s_i \qquad i \leq j$$

$$d_i s_j = \begin{cases} s_{j-1} d_i & i < j \\ \mathrm{Id} & i = j, j+1 \\ s_j d_{i-1} & i > j+1 \end{cases}$$

Diagrammatically X can be depicted as



Def: A simplicial set / $\mathbb{K}$-module / ring / $\mathbb{K}$-algebra etc. is a simplicial object in Set / $\mathbb{K}$-Mod / Ring / $\mathbb{K}$-Alg.

Eg. :) For each $[n] \in \text{obj}\,\Delta$

$\Delta^n := \underset{\Delta}{\text{Hom}}(-, [n])$ is a simplicial set.

In particular, when $n = 0$,

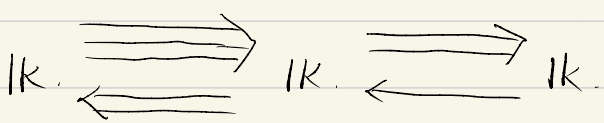
$\Delta^0_k = \underset{\Delta}{\text{Hom}}([k], [0]) = \{*\}$.

We say Δ^0 is a constant simplicial set.

2. Take each Δ^n and post-compose the free functor $\mathbb{k}\Delta^n : \Delta^{op} \xrightarrow{\Delta^n} \text{Set} \xrightarrow{\text{Free}} \mathbb{k}\text{-Mod}$.

we get the simplicial $\mathbb{k}$-module $\mathbb{k}\Delta^n$.

3. Any commutative ring $\mathbb{k}$ can be viewed as the constant simplicial ring $\mathbb{k}\Delta^0$



where all the face and degeneracy maps are $\text{Id}_{\mathbb{k}}$.

Motivation.

1. Simplicial commutative rings have appeared naturally in the study of derived algebraic geometry. Roughly speaking, an affine scheme is a functor

$$\mathrm{Comm}\,\mathbb{k}\text{-Alg} \to \mathrm{Set} \qquad B \mapsto \mathrm{Hom}_{\mathbb{k}\text{-Alg}}(A, B)$$

shall be lifted to

$$\mathrm{s.Comm}\,\mathbb{k}\text{-Alg} \to \mathrm{s.Set} \qquad B \mapsto \overset{h}{\mathrm{Hom}}_{\mathrm{s}\,\mathbb{k}\text{-Alg}}(A, B)$$

which represents a derived affine scheme.

(Töen & Vezzosi 08, Pridham 10, Töen 12)

2. The Dold-Kan correspondence is an equivalence of abelian categories

$$s.\,\mathbb{k}\text{-Mod} \underset{\Gamma}{\overset{N}{\rightleftarrows}} Ch_{\geq 0}(\mathbb{k}\text{-Mod})$$

When restricting to monoids

$$s\,\mathbb{k}\text{-Alg} \underset{\Gamma}{\overset{N}{\rightleftarrows}} \text{connective dg algebra}$$

both functors are only lax monoidal

To study the ring-theoretical properties

(simplicity. separability. Morita theory.

Picard group. Brauer group).

may help us to understand the connection between simplicial algebras and dg algebras.

Also it is interesting to study how the ring / algebra structure constrain the simplicial structure.

3. Reyes and Rogalski (22) showed that twisted Calabi-Yau algebra of dimension 0 are exactly the separable algebras.

To study separable simplicial algebras may help us to generalise and characterize.

twisted Calabi-Yau monoids in suitable monoidal categories

Separability

Let R be a commutative simplicial ring.

(commutative monoids in sAb)

Define simplicial R-modules as module objects over R in the usual way.

Then $sRMod$ is closed symmetric monoidal category (Quillen 67 and Schwede & Shipley 00)

degreewise tensor product

$$(M \underset{R}{\otimes} N)_n := M_n \underset{R_n}{\otimes} N_n$$

unit R.

internal hom $\underline{\mathrm{Hom}}_R(M, N)_n := \mathrm{Hom}_{sR\mathrm{Mod}}(R\Delta^n \otimes_R M, N)$

for $R\Delta^n = R \underset{\mathbb{Z}}{\otimes} \mathbb{Z}\Delta^n$.

Def: A simplicial R-algebra A is

a monoid object in $sR\mathrm{Mod}$.

Let A be a simplicial R-algebra.

$A^e = A \underset{R}{\otimes} A^{op}$ be the enveloping algebra.

Then A is a left simplicial A^e-module under action $(a \otimes b) \cdot x = axb$.

$\forall\ a, x, b \in A_n$. There is a surjective left simplicial A^e-module morphism.

$$\mu_{A/R} : A^e \to A \qquad u \mapsto u \cdot 1_{A_n}$$

$\forall\ u \in (A^e)_n$.

Def: A simplicial R-algebra A is separable over R if $\mu_{A/R}$ splits.

This agrees with the notion of a separable monoids in monoidal category (Pareigis 76). braided monoidal category (Van Oystaeyen & Zhang 98)

Prop (HSZZ)

Let $R = \mathbb{k}$ be a constant commutative simplicial ring and A be a simplicial R-algebra. Then A is separable over R iff $A \cong A_0 \Delta^{\cdot}$ is a constant simplicial ring and A_0 is separable over $R = \mathbb{k}$.

This show separability forces the simplicial structure to be very rigid for simplicial algebras over a constant simplicial ring

What about simplicial algebras over a nonconstant simplicial ring ?

Def: Let A be simplicial $\mathbb{k}$-algebra and define the simplicial matrix ring

$$M_n(A) : \Delta^{op} \xrightarrow{A} \mathbb{k}\text{-Alg} \xrightarrow{M_n(-)} \mathbb{k}\text{-Alg}$$

When A is commutative.

$M_n(A)$ is a simplicial A-algebra.

Prop: (HSZZ)

Let A be a simplicial $\mathbb{K}$-algebra.

1. $M_n(A)$ is separable over $\mathbb{K}$.

iff $M_n(A) \simeq M_n(A_0) A^\circ$
and $M_n(A_0)$ separable over $\mathbb{K}$.

iff $A \simeq A_0 A^\circ$
and A_0 separable over $\mathbb{K}$.

iff A separable over $\mathbb{K}$.

2. If A is commutative.

$M_n(A)$ is separable over A.

When A is nonstant commutative.

the rigidity fails to hold for simplicial

algebras over nonconstant simplicial ring.

Brauer groups.

Classically. an Azumaya R-algebra A

can be defined by

- A central separable over R.
- A^e Morita equivalent to R.

Pareigis 76. Van Oystaeyen & Zhang 98

Borceux & Vitale 02 failed to unify

the conditions in general monoidal categories.

Let A be a simplicial R-algebra.

Then sAMod is enriched over sRMod.

(Hovey 99. Lewis & Mandell 01)

Def. We say two simplicial R-algebras

A and B are simplicial Morita equivalent

if sAMod and sBMod are equivalent

as sRMod-enriched categories

Prop (HSZZ) Let A be a simplicial R-algebra. Then A is central separable over R iff A^e is simplicial Morita equivalent to R.

Theorem (HSZZ) Let A be a simplicial R-algebra.

1. A is separable over R iff A is Azumaya over $Z(A)$

and $Z(A)$ is separable over R.

2) A is Azumaya over R

iff $A \simeq R \otimes_{R_0} A_0$ and

A_0 is Azumaya over R_0

Def: Let R be a commutative simplicial ring

Define $B(R)$ = isoclasses of Azumaya

simplicial R-algebras and

$Br(R) = B(R)/\sim$ where

$[A] \sim [B]$ if A is simplicial

Morita equivalent to B.

Prop (HSZ) $Br(R) \simeq Br(R_0)$

$$[A] \mapsto [A_0]$$

$$[R \otimes_{R_0} T] \leftarrow [T]$$

Future Questions:

1. How define a simplicial abelian group $\underline{Br}(R)$ where $\underline{Br}(R)_0 = Br(R)$?

2. Is twisted Calabi-Yau simplicial algebra equivalent to separable simplicial algebras ?